You have been supplied with the following source files:

`HashTableData.txt`

Open the evidence document, **evidence.doc**

Make sure that your name, centre number and candidate number will appear on every page of this document. This document must contain your answers to each question.

Save this evidence document in your work area as:

evidence_ followed by your centre number_candidate number, for example: **evidence_zz999_9999**

A class declaration can be used to declare a record. If the programming language used does **not** support arrays, a list can be used instead.

One source file is used to answer **Question 3**. The file is called **`HashTableData.txt`**

1 A program stores integers in a stack. The stack is represented as a 1D array of 30 elements with the identifier `Stack`

The global pointer `TopOfStack` stores the index of the last element inserted into the stack. `TopOfStack` is initialised to `−1`

(a) Write program code to declare `Stack`, initialise each element in the array with a null value and declare and initialise `TopOfStack`

Save your program as **Question1_N25**. Copy and paste the program code into part **1(a)** in the evidence document.

[2]

(b) The function `Push()` takes an integer parameter. If the stack is full, the function returns `FALSE`. If the stack is not full, the parameter is inserted into the stack, the pointer is updated and the function returns `TRUE`

Write the program code for `Push()`

Save your program. Copy and paste the program code into part **1(b)** in the evidence document.

[4]